


```

    }
    printf ("%i. d", c[i]);
}
printf ("\n");
}

int main()
{
    int a1[] = {1, 2, 3, 4, 5, 5, 6, 6};
    int a2[] = {3, 4, 5, 6, 7, 7};
    int size1 = sizeof(a1);
    int size2 = sizeof(a2);
    findCommonElements(a1, size1, a2, size2);
}

```

Q2)

```

#include <stdio.h>

void readMatrix(int m[10][10], int r, int c)
{
    printf ("Enter rows & columns\n");
    for (int i = 0; i < r; i++)
        printf ("Enter elements in array: \n");
    for (int i = 0; i < r; i++)
        for (int j = 0; j < c; j++)
            scanf ("%i. d", &m[i][j]);
}

void multiplyMatrices(int m1[10][10], int m2[10][10],
    int r1, int c1,
    int r2, int c2)
{

```

```
for (int i = 0; i < r1; i++)  
{  
    for (int j = 0; j < c1; j++)  
    {  
        result[i][j] = 0;  
    }  
}  
  
for (int i = 0; i < r1; i++)  
{  
    for (int j = 0; j < c2; j++)  
    {  
        for (int k = 0; k < c1; k++)  
        {  
            result[i][j] += m1[i][k] * m2[k][j];  
        }  
    }  
}  
  
void displayMatrix (int m1[10][10], int r, int c)  
{  
    printf ("Resulting Matrix: \n");  
    for (int i = 0; i < r; i++)  
    {  
        for (int j = 0; j < c; j++)  
        {  
            printf ("%d", m1[i][j]);  
        }  
        printf ("\n");  
    }  
}  
  
int main ()
```

§
 int m1[10][10], m2[10][10], result[10][10];
 int r1, c1, r2, c2;
 printf ("Enter no. of rows & columns :");
 scanf ("%d", &m1[r1][c1]); for matrix 1
 printf ("Enter no. of rows & columns for matrix 2 :");
 scanf ("%d", &r2, &c2);
 if (c1 != r2)

printf ("Matrix Multiplication not possible :");
 §

readMatrix (m1, r1, c1);
 readMatrix (m2, r2, c2);
 multiplyMatrices (m1, m2, result, r1, c1, r2, c2);
 displayMatrix (m1, r1, c2);
 §

Q3)

#include <stdio.h>
 #include <string.h>
 int areAnagrams (char s1[], char s2[])

§
 int count[26] = {0};
 for (int i = 0; s1[i] != '\0'; i++)

§
 if (s1[i] != ' ')

count[s1[i] - 'a']++;
 § §

for (int i = 0; s2[i] != '\0'; i++)

§
 if (s2[i] != ' ')

```

Q4) #include <stdio.h>
float add ( float a, float b)
{
    return a+b;
}
float subtract ( float a, float b)
{
    return a-b;
}
float multiply ( float a, float b)
{
    return a * b;
}
float divide ( float a, float b)
{
    if ( b == 0)
    {
        printf ( "Error\n" );
    }
    else
    {
        return a/b;
    }
}
int main ()
{
    float num1, num2;
    int choice;
    float *operation ( float, float );
    printf ( "Enter first no.\n" );
    scanf ( "%f", &num1 );
    printf ( "Enter second no : \n" );

```

```

scanf ("%f %d" & num1, & num2);
printf ("Select an operation");
printf ("1: Addition\n");
printf ("2: Subtraction\n");
printf ("3: Multiplication\n");
printf ("4: Division\n");
printf ("Enter your choice");
scanf ("%d", & choice);
switch (choice)
{

```

case 1:

```

operation = add;
break;

```

case 2:

```

operation = subtraction;
break;

```

→ case 3:

```

operation = multiply;
break;

```

~~op~~ case 4:

```

operation = divide;
break;

```

default:

```

printf ("Invalid choice");
}

```

```

float result = operation (num1, num2);
if (choice == 4 && num2 == 0)
{

```

```

printf ("Error");
}

```

else

{

```

printf ("Result: %.2f\n", result);
}
}

```

Q5) #include <stdio.h>
void sortDescending (int scores[], int n)
{
for (int i = 0; i < n-1; i++)
{
for (int j = 0; j < n-i-1; j++)
{
if (scores[j] > scores[j+1])
{
int temp = scores[j];
scores[j] = scores[j+1];
scores[j+1] = temp;
}
}
}
}

void sortDescending (int scores[], int n)
{
for (int i = 0; i < n-1; i++)
{
for (int j = 0; j < n-i-1; j++)
{
if (scores[j] < scores[j+1])
{
int temp = scores[j];
scores[j] = scores[j+1];
scores[j+1] = temp;
}
}
}
}

```
void displayScores (int scores[], int n)
{
    printf ("Sorted Scores:\n");
    for (int i = 0; i < n; i++)
    {
        printf ("%d", & scores[i]);
    }
    printf ("\n");
}

int main()
{
    int scores [100];
    int n, choice;
    printf ("Enter the no. of scores:\n");
    scanf ("%d", &n);
    printf ("Enter the scores\n");
    for (int i = 0; i < n; i++)
    {
        scanf ("%d", & scores[i]);
    }

    void (* sortFunction) (int[], int);
    printf ("Choose sorting order");
    printf ("1. Ascending\n");
    printf ("2. Descending\n");
    printf ("Enter your choice");
    scanf ("%d", & choice);
    switch (choice)
    {
        case 1:
            sortFunction = sortAscending;
            break;
        case 2:
```

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```

sortFunction = sortDescending;
break;
default:
printf("Invalid choice");
}

sortFunction(scores, n);
displayScores(scores, n);
}

```

86) #include <stdio.h>

```

void toUpperCase(char str[])
{
    for (int i=0; str[i] != '\0'; i++)
    {
        str[i] = toupper(str[i]);
    }
}

void toLowerCase(char str[])
{
    for (int i=0; str[i] != '\0'; i++)
    {
        str[i] = tolower(str[i]);
    }
}

int main()
{
    char str[100], man[100];
    int choice;
    printf("Enter a string\n");
    fgets(str, 100, stdin);
    void (*convertString)(char[]);
}

```

```
printf ("Enter your choice:");  
scanf ("%d", &choice);  
if (choice == 1)  
{  
    convertString = toUpperCase;  
}  
else if (choice == 2)  
{  
    convertString = toLowerCase;  
}  
else  
{  
    printf ("Invalid Choice");  
}  
convertString (str);  
printf ("Modified String: %s", str);
```

Q8)

```
#include <stdio.h>  
# double areaOfCircle (double radius)  
{  
    return 3.14 * radius * radius;  
}  
double areaOfSquare (double side)  
{  
    return side * side;  
}  
double areaOfRectangle (double length, double width)  
{  
    return length * width;  
}
```

```

}
int main ()
{
    int choice;
    double area;
    double (*areaFunction)(double);
    printf ("Shape Area Calculator");
    printf ("1. Circle \n");
    printf ("2. Rectangle \n");
    printf ("3. Square \n");
    printf ("Choose a shape");
    scanf ("%d", &choice);
    switch (choice)
    {

```

```

        case 1:

```

```

    {

```

```

        double radius;
        printf ("Enter radius of circle");
        scanf ("%lf", &radius);
        areaFunction = areaOfCircle;
        area = areaFunction (radius);
        break;
    }

```

```

        case 2:

```

```

    {

```

```

        double side;
        printf ("Enter the side length of square:");
        scanf ("%d", &side);
        areaFunction = areaOfSquare;
        area = areaFunction (side);
        break;
    }
}

```

case 3:

{

double length, width;

printf ("Enter the length & width of the
scanf ("%lf %lf", &length, &width);

area = areaOfRectangle (length, width);

areaFunction = areaOfRectangle;

break;

}

default:

printf ("Invalid Choice");

}

printf ("The area is: %lf\n", area);

}